

Chapter 06 Homework Answer Sheet

Name: _____ Student ID: _____ Instructor: _____ Score: _____

Part A multiple choice (15 points)

1-5 BBAEA

6-10 DDDEC

11-15 ACEBC

Part B short question (15 points)

16. (5 points)

(a) (1 point) $n_f = 1$ would produce a photon with the shortest wavelength.

(b) (3 points)

$$\Delta E = (-2.18 \times 10^{-18}) \times (1/n_f^2 - 1/n_i^2) = (-2.18 \times 10^{-18}) \times (1/1^2 - 1/4^2) = -2.04 \times 10^{-18} \text{ J}$$

energy of such a photon:

$$E_{\text{photon}} = -\Delta E = \underline{2.04 \times 10^{-18} \text{ J}}$$

wavelength of such a photon:

$$\lambda = hc/E_{\text{photon}} = (6.626 \times 10^{-34} \times 2.998 \times 10^8) / ((2.18 \times 10^{-18}) \times (1/1^2 - 1/4^2)) = \underline{9.72 \times 10^{-8} \text{ m}} \quad \text{OR} \quad \underline{97.2 \text{ nm}}$$

The electromagnetic spectrum lies in Ultra-violet region.

(c) (2 points) The image on the right shows the output of a detector that measures the intensity of light emitted from a sample of hydrogen atoms that have been excited so that each atom begins with an electron in the $n = 4$ state. What is the final state, n_f , of the transition being detected?

$$E_{\text{photon}} = hc/\lambda = (6.626 \times 10^{-34} \times 2.998 \times 10^8) / (4.86 \times 10^{-7}) = 4.09 \times 10^{-19} \text{ J}$$

$$\Delta E = -4.09 \times 10^{-19} \text{ J}$$

$$\Delta E = (-2.18 \times 10^{-18}) \times (1/n_f^2 - 1/n_i^2) = -4.09 \times 10^{-19} \text{ J}$$

$$1/n_f^2 - 1/n_i^2 = 0.187$$

$$1/n_f^2 - 1/4^2 = 0.187$$

$$n_f \approx 2$$

17. (4 points)

$$(a) (1 \text{ point}) E = h\nu = 6.62 \times 10^{-34} \text{ J}\cdot\text{s} \times 1.09 \times 10^{15} \text{ s}^{-1} = 7.22 \times 10^{-19} \text{ J}$$

The minimum energy needed to eject an electron is $7.22 \times 10^{-19} \text{ J}$.

$$(b) (1 \text{ point}) \lambda = c/\nu = 3.00 \times 10^8 \text{ m}\cdot\text{s}^{-1} \div (1.09 \times 10^{15} \text{ s}^{-1}) = 2.75 \times 10^{-7} \text{ m} \text{ OR } 275 \text{ nm}$$

Radiation with 275 nm will provide a photon of this energy

(c) (1 points)

$$E_{120} = hc/\lambda = 6.62 \times 10^{-34} \text{ J}\cdot\text{s} \times 3.00 \times 10^8 \text{ m}\cdot\text{s}^{-1} \div (120 \times 10^{-9} \text{ m}) = 1.66 \times 10^{-18} \text{ J}$$

$$E_k = E_{120} - E = 1.66 \times 10^{-18} \text{ J} - 7.22 \times 10^{-19} \text{ J} = 9.33 \times 10^{-19} \text{ J} = 9.3 \times 10^{-19} \text{ J}$$

The maximum possible kinetic energy of the emitted electrons is $9.3 \times 10^{-19} \text{ J}$.

18. (6 points)

(a), (b)

(1) F : [He]2s ² 2p ⁵	one unpaired electron	1
(2) Zn : [Ar]4s ² 3d ¹⁰	zero unpaired electron	0
(3) Si : [Ne] 3s ² 3p ²	two unpaired electrons	2